

■ FEATURES OF 1GR-FE ENGINE

The 1GR-FE engine has been able to achieve the following performance through the adoption of the items listed below.

- (1) High performance and reliability
- (2) Low noise and vibration
- (3) Lightweight and compact design
- (4) Good serviceability
- (5) Clean emission and fuel economy

Item		(1)	(2)	(3)	(4)	(5)
Engine Proper	An upright intake port is used.	○		○		
	A taper squish shape is used for combustion chamber.	○				○
	A cylinder block made of aluminum alloy is used			○		
	The piston skirt is coated with resin.	○	○			
	An oil pan No.1 sub-assembly made of aluminum alloy is used.		○	○		
Valve Mechanism	The shim-less type valve lifters are used.	○		○		
	Timing chains and chain tensioners are used.	○	○	○		
Lubrication System	A water-cooled type engine oil cooler is used.	○				
Cooling System	TOYOTA Genuine SLLC (Super Long Life Coolant) is used as the engine coolant.				○	
Intake and Exhaust System	The intake air chamber made of plastic is used.			○		
	The link-less type throttle body is used.			○	○	
	Stainless steel exhaust manifolds are used.			○		
	Ceramic type TWCs (Three Way catalytic Converters) are used.					○
Fuel System	12-hole type fuel injectors are used to improve the atomization of fuel.	○				○
	The fuel delivery pipe made of plastic is used.			○		
Ignition System	The DIS (Direct Ignition System) makes ignition timing adjustment unnecessary.	○			○	○
	The long-reach type spark plugs is used.	○				
Charging System	Segment Conductor type generator is used.	○		○		
Starting System	PS (Planetary reduction-Segment conductor motor) type starter is used.			○		
Serpentine Belt Drive System	A serpentine belt drive system is used.			○	○	

(Continued)

Item		(1)	(2)	(3)	(4)	(5)
Engine Control System	MRE (Magnetic Resistance Element) type VVT sensors are used.	☐				
	The ETCS-i (Electronic Throttle Control System- intelligent) is used.	☐				☐
	The VVT-i (Variable Valve Timing-intelligent) system is used.	☐				☐
	The ACIS (Acoustic Control Induction System) is used.	☐				☐
	The cranking hold function is used.	☐				